

On the genetic diversity of spiny mice (genus *Acomys*) and gerbils (genus *Gerbillus*) in the Arabian Peninsula

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Using non-destructive sampling we provide further genetic characterisations for spiny mice (*Acomys dimidiatus/cahirinus*) and gerbils (*Gerbillus* sp.) in three regions in Saudi Arabia. All individuals were sequenced for a fragment of the cytochrome *b* gene, and compared against available conspecifics and closely related taxa. We confirm the existence of a second *Acomys dimidiatus/cahirinus* lineage specific to the Arabian Peninsula as seen previously. The Arabian *Gerbillus nanus* is shown to group with Middle Eastern rather than African conspecifics. A second cryptic *Gerbillus* lineage was also sampled across multiple locations, which may be an uncharacterised *G. dasyurus*.

Keywords: *Acomys*; genetic diversity; *Gerbillus*; rodents; Saudi Arabia.

Introduction

Partially due to its diversity of continental connections, the fauna of the Arabian Peninsula reflects a mixture of influences, the majority of studies describing Afro-Arabian affinities across different taxonomic groups (e.g. Pook, Joger, Stumpel, & Wuster, 2009). In line with predictions of high rodent diversity in arid regions (Abramsky & Rosenzweig, 1984), Arabian rodents include species of *Acomys*, *Eliomys*, *Gerbillus*, *Jaculus*, *Meriones*, and *Psammomys*. Among these taxa, the gerbil and spiny mouse are commonly found throughout the Arabian Peninsula, representing genera of Afro-Indian distribution. Studies on spiny mouse (*Acomys*) have included limited sampling in Arabia (Frynta et al., 2010), and some species of gerbil (*Gerbillus*) are as yet unrepresented in GenBank for gene regions of phylogenetic comparison (e.g. *Gerbillus dasyurus*) despite extensive recent work on African representatives (Ndiaye, Shanas, & Granjon, 2013). Here we sample *Acomys* and *Gerbillus* across three Saudi Arabian locations each to give a preliminary indication of their geographical breaks and diversity in the Arabian Peninsula. Specifically we ask if there is a common trend in distribution of genetic variation across the taxonomic groups investigated, and how this relates to the same species in neighbouring regions.

Material and Methods

Gerbils (n=17) and Spiny Mice (n=16) were trapped and non-destructively sampled from three regions in the Kingdom of Saudi Arabia in March-June 2012. Locations sampled for both groups included Abha, Raydah Protected Area, and Taif areas with additional *Acomys* from Riyadh region and gerbils from Hawtat Bani Tamim (coordinates in Appendix 1).

DNA extraction was performed from earlobe tissue using Qiagen extraction reagents and separate spin columns (Epoch Life Science). The mitochondrial cytochrome *b* (cytb) gene was amplified using the primers ALLROD/H15149 (Irwin, Kocher, & Wilson, 1991) for gerbils, and for the spiny mice MOLCIT/MVZ (Ibáñez, García-Mudarra, Ruedi, Stadelmann, & Juste, 2006; Smith &

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Patton, 1993). A shorter spiny mouse fragment was amplified with MTCBF/H15149 (Naidu, Fitak, Munguia-Vega, Culver, 2012; Irwin et al., 1991; all cytb was amplified at 50°C for 40 cycles). A subgroup of gerbils was additionally sequenced for a fragment of the cytochrome oxidase c subunit I (COI) using universal primers LCO1490/HCO2198 (Folmer, Black, Hoeh, Lutz, & Vrijenhoek, 1994), and a fragment of the 16s ribosomal gene (16s) using universal primers COIF/COIR (Palumbi & Benzie, 1991; 40 cycles at 55°C). Denaturation and extension temperatures for all primer combinations were 95 and 72°C respectively. All sequencing was performed on an ABI 3130 sequencing platform.

Sequence data was cleaned and aligned in BIOEDIT (Hall, 2005) and combined with samples available on GenBank (accession numbers in Appendix 1). Nucleotide substitutions and translation into amino acids was performed through MEGA5 (Tamura et al., 2011). Maximum parsimony haplotype networks were calculated using the TCS (Clement, Posada, & Crandall, 2000) application. Model determination and construction of neighbour joining trees were conducted between gene regions separately (1000 bootstraps) using genetic distances inferred under the Tamura and Nei (1993) model for nucleotide sequences, in MEGA5. In this approach a maximum likelihood approach was used to produce phylogenetic trees showing support values for nodes with >80% bootstrap support. Where two genes were available MRBAYES 3.1 (Ronquist & Huelsenbeck, 2003) was used. We used the default models for our gene partitions, setting MrBayes to infer coding bias assuming that only variable characters can be observed for both (“CODING = VARIABLE”). Bayesian analyses were undertaken using four independent runs, each using a random starting tree and 1×10^6 generations with one cold and three heated chains, sampling trees every 100 generations. The trees are derived from a consensus of the last 9901 trees for the first of the four independent runs (removing the initial 100).

Results

Spiny Mouse *Acomys* sp.

A total of 800 base pairs (bp) of cytb sequence data were generated across 16 spiny mice, with a further 29 sequenced for a shorter 404bp fragment. Of these longer 16 sequences; 10 were unique with sequence variation comprising 36 variable bases, 22 of these being parsimony informative. The GTR+G model of sequence evolution was determined to be the best fit to the data (AIC= 4845). Initially a neighbor joining phylogeny was constructed for the greater *A. cahirinus/dimidiatus* species group, but this was reduced to exclude the ‘*A. minous* branch’ to concentrate on the sampled lineages (Figure 1a). A haplotype network with this reduced *A. cahirinus/dimidiatus* set from GenBank identified 19 haplotypes and a split with a minimum distance of 16 substitutions between the southern Saudi Arabian samples and the next closest haplotype (Figure 1b).

Gerbils *Gerbillus* sp.

For cytb 404bp were generated for 18 individuals, giving 13 unique sequences. Sequence variation comprised 70 variable bases, 53 of these being parsimony informative. The GTR+G model of sequence evolution was determined to be the best fit to the data (AIC= 3513). Both the neighbor joining phylogeny (Figure 2a), and haplotype network (Figure 2b) showed that of the sampled *Gerbillus*, nine individuals fell within the GenBank representatives of *G. nanus*, and a further eight individuals (of mixed localities) fell outside this group.

To further explore the discrepancy between the deeper split within the Saudi Arabian *Gerbillus* samples two additional genes were sequenced across this split; seven unique 656bp COI fragments (n=7) and 3 unique 570bp 16s sequences (n=7) were re-

covered. A three-gene combined MRBAYES consensus tree provided a well supported divergence between these lineages (Appendix 3).

Discussion

Tilman's prediction of high desert diversity has been shown to hold true at least in rodents (Abramsky & Rosenzweig, 1984), suggesting that arid environments are particularly important for small mammal species assemblages. Most small mammal species in Saudi Arabia are genetically represented by only a small number of individuals from a single location. By sampling three small mammals from multiple locations we greatly increase the available knowledge of these species in this region, as can be seen with just a cursory glance at the figures presented here.

Within *G. nanus* there appears to be a break at the western edge of the Arabian Peninsula as shown in Ndiaye et al. (2013), with Libyan specimens grouping alongside those further west. The Saudi Arabian *G. nanus* groups more closely with haplotypes from Israel as well as the haplotype available from Pakistan, in line with this previously recognised split. Of interest here is that of the Saudi Arabian *Gerbillus* sampled was a lineage clearly divergent from the other *G. nanus*. This clade was present in both the southern and western sampling locations (although there is divergence between locations). What is currently unclear is whether this divergent lineage represents the currently genetically uncharacterised *G. dasyurus*. Further combined morphological and genetic determination will be necessary to clarify this issue. If there is regional ecological differentiation, as suggested for the spiny mouse, the absence of the recognised *G. nanus* in the south might represent its replacement in this area. More exhaustive sampling is necessary to better confirm species ranges in these species.

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Appendix

Appendix 1. Accession numbers for all sequence data collected during this study.

Appendix 2. References for additional samples used as described in appendix 2.

Appendix 3. A combined consensus Bayesian phylogeny for *Gerbillus* in Saudi Arabia using cytb, COI, and 16s showing support values for each node.

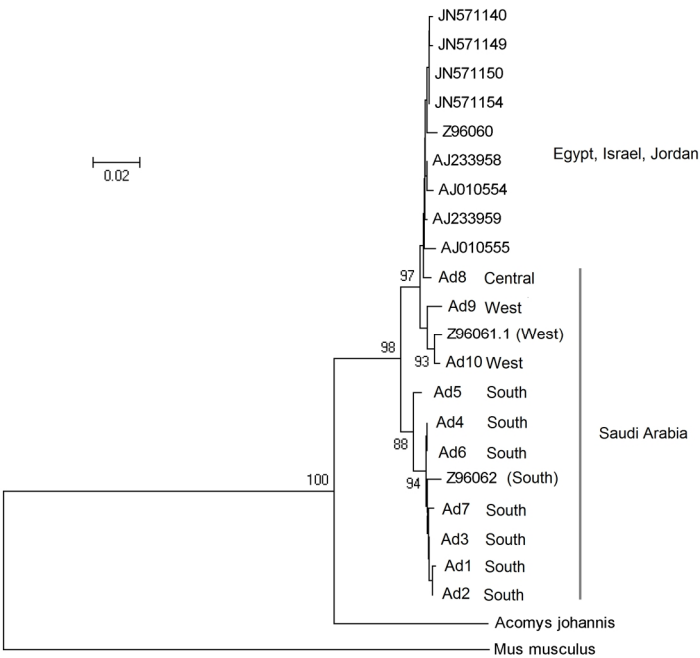


Figure 1a. A cytochrome b neighbour joining phylogeny for *Acomys cahirinus/dimidiatus* (a GTR+G model of sequence evolution was used, bootstrap support values >80% presented).

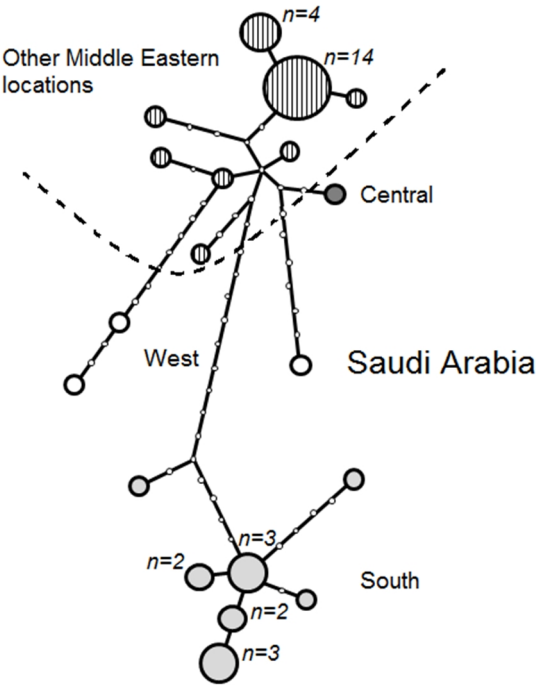


Figure 1b. Maximum parsimony haplotype network for *Acomys dimidiatus* showing numbers of individuals for each haplotype and numbers of mutational steps between them (small circles represent single steps).

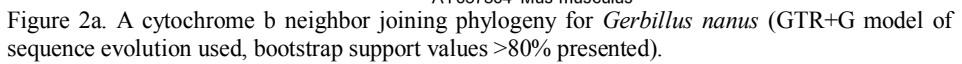
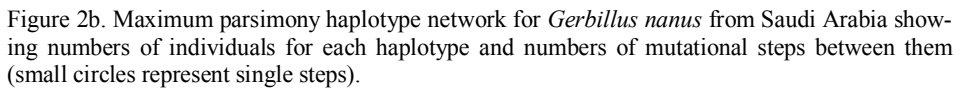


Figure 2a. A cytochrome b neighbor joining phylogeny for *Gerbillus nanus* (GTR+G model of sequence evolution used, bootstrap support values >80% presented).



Appendix 1

GENBANK accession numbers and sequence details (additional references in Appendix 2)

Cytochrome b gene

Species	N	Origin	GenBank	Reference
<i>Acomys dimidiatus</i>	3	South. Ad1 Raydah, Saudi Arabia 18°02' N, 42°50'E	KF422682	This study
<i>A. dimidiatus</i>	2	South. Ad2 Raydah, Saudi Arabia 18°02'N, 42°50'E	KF422683	This study
<i>A. dimidiatus</i>	3	South. Ad3 Tamniah, Saudi Arabia 18°01'N, 42°46'E	KF422684	This study
<i>A. dimidiatus</i>	1	South. Ad4 Raydah, Saudi Arabia 18°02' N, 42°50'E	KF422685	This study
<i>A. dimidiatus</i>	1	South. Ad5 Raydah, Saudi Arabia 18°02' N, 42°50'E	KF422686	This study
<i>A. dimidiatus</i>	1	South. Ad6 Raydah, Saudi Arabia 18°02' N, 42°50'E	KF422687	This study
<i>A. dimidiatus</i>	1	South. Ad7 Abha, Saudi Arabia 18°02' N, 42°50'E	KF422688	This study
<i>A. dimidiatus</i>	1	Central. Ad8 Thumama, Saudi Arabia 25°13' N, 46°38'E	KF422689	This study
<i>A. dimidiatus</i>	2	West. Ad9 Taif, Saudi Arabia 21°16' N, 40°41'E	KF422690	This study
<i>A. dimidiatus</i>	1	West. Ad10 Taif, Saudi Arabia 21°16' N, 40°41'E	KF422691	This study
<i>A. dimidiatus</i>		West. Taif, Saudi Arabia 21°16' N, 40°41'E	Z96061	Barome et al. 1998
<i>A. johannis</i>		Cameroon	HM635822	Dobigny et al. 2011
<i>A. dimidiatus</i>	4	(A101) Israel 35°03'N, 3243.8'E	JN571140	Hadid unpublished
<i>A. dimidiatus</i>	13	(E22) Israel 35°03'N, 3243.8'E	JN571150	Hadid unpublished

<i>Gerbillus nanus</i>	3	West. G1 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422692	This study
<i>Gerbillus nanus.</i>	2	West. G2 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422693	This study
<i>Gerbillus</i> sp.	2	West. G3 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422694	This study
<i>Gerbillus nanus</i>	2	West. G4 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422695	This study
<i>Gerbillus nanus</i>	1	West. G5 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422696	This study
<i>Gerbillus nanus</i>	1	West. G6 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422697	This study
<i>Gerbillus nanus</i>	1	West. G7 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422698	This study
<i>Gerbillus nanus</i>	1	Central. G8 Hawtat, Saudi Arabia 23°17'N, 46°32'E	KF422699	This study
<i>Gerbillus</i> sp.	1	South. G9 Raydah, Saudi Arabia 18°02'N, 42°50'E	KF422700	This study
<i>Gerbillus</i> sp.	1	South. G10 Habala, Saudi Arabia 18°02'N, 42°50'E	KF422701	This study
<i>Gerbillus</i> sp.	1	South. G11 Habala, Saudi Arabia 18°02'N, 42°50'E	KF422702	This study
<i>Gerbillus</i> sp.	1	South. G12 Abha, Saudi Arabia 18°02'N, 42°50'E	KF422703	This study
<i>Gerbillus</i> sp.	1	South. G13 Tamniah, Saudi Arabia 18°01'N, 42°46'E	KF422704	This study
<i>Gerbillus henleyi</i>	1	Niger	JQ753050	Ndiaye <i>et al.</i> 2012
<i>Gerbillus nanus</i>	13	Mali - Pakistan	JQ753051-63	Ndiaye <i>et al.</i> 2012
<i>Mus musculus</i>		-	AY057804	Lundrigan et al. 2002
<i>Eliomys melanurus</i>	1	South. Raydah, Saudi Arabia 18°02'N, 42°50'E	KF422705	This study
<i>Mus sp</i>	1	South. Raydah, Saudi Arabia 18°02'N, 42°50'E	KF422706	This study
<i>Mus sp</i>	1	South. Raydah, Saudi Arabia 18°02'N, 42°50'E	KF422707	This study

Cytochrome Oxidase I (COI) gene

Species	N	Origin	GenBank	Reference
<i>Gerbillus nanus</i>	1	West. COI1 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422708	This study
<i>Gerbillus nanus</i>	1	West. COI2 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422709	This study
<i>Gerbillus nanus</i>	1	West. COI3 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422710	This study
<i>Gerbillus sp.</i>	1	West. COI4 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422711	This study
<i>Gerbillus sp.</i>	1	West. COI5 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422712	This study
<i>Gerbillus nanus</i>	1	West. COI6 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422713	This study
<i>Gerbillus nanus</i>	1	West. COI7 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422714	This study
<i>Meriones libycus</i>		COIMI Thumama, Saudi Arabia 25°13'N, 46°38'E	KF422715	This study
<i>Mus musculus</i>		-	AB444046	Nakamura et al. unpublished

16s Ribosomal gene

Species	N	Origin	GenBank	Reference
<i>Gerbillus sp.</i>	2	West. G161 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422716	This study
<i>Gerbillus nanus</i>	3	West. G162 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422717	This study
<i>Gerbillus nanus</i>	1	West. G163 Taif, Saudi Arabia 21°16'N, 40°41'E	KF422718	This study
<i>Meriones libycus</i>		M11 Thumama, Saudi Arabia 25°13'N, 46°38'E	KF422719	This study
<i>Gerbillus nigeriae</i>		Mauritania	AF141257	Ducroz et al. 2001
<i>Mus musculus</i>		-	AP013031	Imanishi et al. unpublished

Appendix 2

References additional to main text

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